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Crop Recommender: Machine Learning–Based Computational Method to Recommend the Best Crop Using Soil and Environmental Features

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5.1 INTRODUCTION

In our country, agriculture is one of the main sectors on which a large number of people are dependent [1]. In spite of having a great variety of topographic characteristics, landforms, and climatic zones that support various types of cultivation, the yield obtained is still less due to field mismanagement and the erroneous selection of crops [2]. To adequately nourish the rapidly expanding population, it is crucial to prioritize the proportional increase in crop yield. This can be accomplished by addressing the underlying causes of reduced crop production, including insufficient planning, mismanagement of livestock, and the unpredictable nature of weather conditions. Since the complicated phenomenon of crop production is impacted by agro-climatic input parameters, therefore the specifications for agricultural inputs vary according to the farmer and field [3]. In addition to the various factors that affect crop production, another mammoth problem that agriculturists are facing is the rise of global warming. Global warming being on peak leads to droughts and non-predictable climatic conditions. Moreover, farmers working in the traditional ways

lead to the problem of field mismanagement and hence affect productivity. According to the reports, due to unpredictable climatic conditions over the years, the farmer's income has reduced by 20.25% [4].

Agriculture holds a significant position in India's gross domestic product (GDP). The crucial factor for achieving a high crop yield is the proper selection of appropriate crops at the right time of year. However, farmers often rely on market trends and ancestral instincts, which have proven to be ineffective methods [5]. Adapting the proposed methodology will help farmers overcome the problem of scarcely quantifiable yield and contribute more and more toward the country's GDP [6]. The use of algorithms will also aid in reaching an efficient, cost-effective solution to the difficulties faced by the farmers. The algorithms trained and tested will be used for crop recommendation based on various inputs like Nitrogen, Phosphorus, Potassium (NPK) content and pH of soil, temperature, and humidity of the area. The predicted crops included in this methodology are *maize, cotton, rice, coffee, pigeon peas, moth beans, jute, chickpea, coconut, papaya, orange, apple, muskmelon, watermelon, grapes, kidney beans, banana, pomegranate, lentil, black gram, mung bean, and mango*.

Some of the primary novel contributions of this chapter are the following:

- In this study, a system has been proposed that will enable agriculturists to select accurate crops based on environmental factors and soil parameters.
- The proposed system can reduce the erroneous methods applied during the selection of a crop, which in turn can prove helpful in improving crop productivity.
- A crop recommender (CR) tailored to provide recommendations for the most suitable crop to plant will be developed.

Four sections make up this chapter. In the first section, an introduction has been presented, and various issues related to decreased crop yield have been discussed. The second section contains an account of related work done in this field. An experimental scenario has been covered in the third section, which includes the dataset used and methods employed. Results have been provided in the fourth section. Lastly, a conclusion of the study has been drawn.

5.2 RELATED WORK

The work completed in the area of crop recommendation by various researchers is included in this section. A thorough description of the study project is provided in the remaining portion of this section, and the summarization of this section is given in Table 5.1. Doshi et al. [1] developed Agro Consultant, which takes into consideration all the relevant attributes – i.e., temperature, location, rainfall, and soil condition – for predicting crop appropriateness using four algorithms – namely, DT, Random Forest (RF), NN, and K-nearest neighbor. This study also presented a rainfall predictor (a secondary system) that enables early predictions of rainfall [1]. Kumar et al. [7] built a model called a selection of crops, which is based on parameters like weather, soil type, water density, and crop kind. This method employed models like RF, support vector machine (SVM), DT, KNN, and Regularized Greedy Forest (RGF) [7]. Kumar Rajak et al. [8] employed a method that was trained on a soil database obtained from a

TABLE 5.1

Related Work

Author	Year	Description	Algorithm
Doshi et al.	2018	Prediction of crop suitability using all relevant attributes (i.e., temperature, rainfall, and soil condition)	Decision tree, NN, RF, and KNN
Kumar et al.	2015	Selection of crops using parameters soil type, water density, and crop kind	RF, SVM, decision tree, KNN, and RGF
Kumar Rajak et al.	2017	Crop recommendation using a validated dataset for maximizing crop yield	Bagging ANN, SVM, Naive Bayes, RF
Savla et al.	2015	Comparative analysis using various classification algorithms, particularly on soybean crops	SVM, NN, RF, REP Tree
Shakil Ahamed et al.	2015	Data mining techniques were conducted for the cereal crops	KNN, linear regression, ANN
Suresh et al.	2021	Developed a system called efficient crop yield, which predicts the suitability of crops by employing two datasets (i.e., datasets of crop and particular location)	SVM

farm. The data retrieved after testing from the lab was then provided to a recommendation system based on an ensemble model of Artificial Neural Network (ANN) and SVM [8]. Savla et al. [9] made a relative analysis of classification algorithms and analyzed the rate of prediction in precision agriculture. The algorithms employed in the study were RF, NN, SVM, Reduced Error Pruning Tree (REP Tree), Bagging, and Bayes. Among the aforementioned algorithms, the bagging algorithm drew the best result for predicting the yield [9]. Shakil et al. [10] specified the use of data mining techniques that can help government organizations and agriculturists draw out the necessary information to reach better conclusions. This ultimately results in increased production levels. These techniques were particularly applied for estimating cereal crops in Bangladesh. The algorithms employed in their study included clustering and classification using KNN, Linear Regression, and ANN [10]. Suresh et al. [11] developed an “Efficient Crop Yield Recommendation System” based on SVM that recognizes a specific crop in accordance with the given data. The system recommended a specific crop in accordance with the parameters (NPK and pH) values [11].

5.3 DATA AND METHODS

5.3.1 Experimental Scenario

For experimentation, the programming environment python is adopted. Bagging models have been used for experimentation. The steps of the experiment are listed in Figure 5.1.

The steps undertaken in this study are as follows: procure the relevant dataset, preprocess the obtained dataset, visualize the dataset, build the ML model, and evaluate the same. The data is preprocessed and then fed to models for classification. Lastly, the comparative analysis between the ensemble of homogenous classifiers and single classifiers is conducted.

5.3.2 Dataset Collection

The data used for the study is a combination of two datasets procured from Kaggle [12] crop recommendation and *agric dataset*, which consist of 2,202 and 4,100 samples, respectively. After merging the samples, the resultant dataset was divided into 7:3 portions for train and testing.

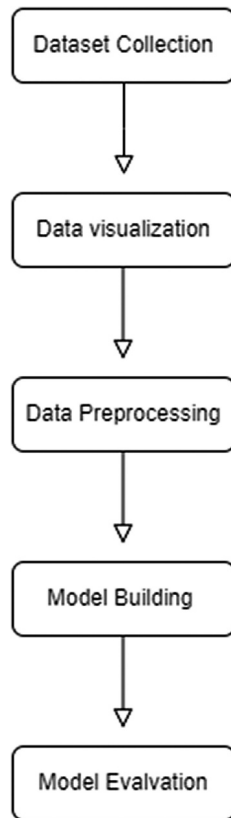


FIGURE 5.1
Experimental steps.

5.3.3 Data Visualization

Data visualization plays a crucial role in simplifying intricate information and making it comprehensible. To assess the correlation among various features, a heat map has been utilized. The heat map indicates a significant positive correlation between phosphorous (P) and potassium (K) while revealing a negative correlation between phosphorous (P) and rainfall, pH and potassium (K), and nitrogen (N) and temperature, as presented in Figure 5.2. “Univariate Analysis.” After comprehending the correlations among the features, this data becomes instrumental in making informed decisions for selecting the most suitable crop. Following the univariate analysis by the bar graph aids in identifying the specific parameters needed for the particular crop, as depicted in Figure 5.3.

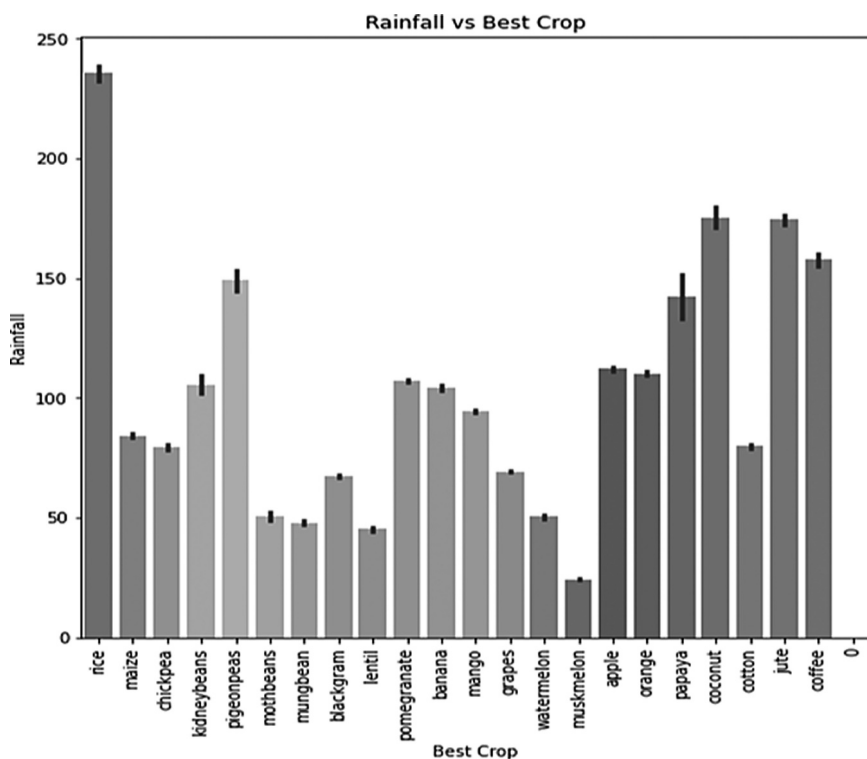


FIGURE 5.2
Univariate analysis.

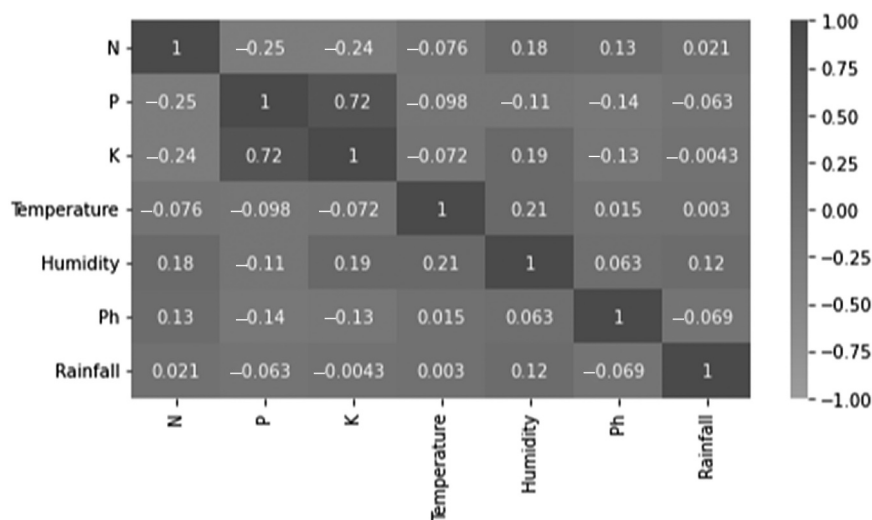


FIGURE 5.3
Heat map.

5.3.4 Data Cleaning

In the discipline of ML, data cleaning (DC) is an essential and necessary step that precedes the modeling process. This method involves identifying and rectifying errors, discrepancies, and inaccuracies present in a dataset. This process encompasses various tasks, including addressing missing values, eliminating duplicates, standardizing data formats, and handling outlier values. Adhering to this step is essential; otherwise, it can result in the creation of an inaccurate model [13].

Allowing the model to learn from pertinent or relevant data ultimately leads to improved classification outcomes [14]. After merging the datasets [12], a few null entries present in the data were removed.

5.3.4.1 Outlier Handling

One of the main parts of data preprocessing is the outlier and noise handling: fault in data collection, error during data-entering, unknown encoding, inconsistent data, inconsistent formats, etc., correspond to different types of noise and outliers [15]. It's very significant to handle it before modeling, as it affects the statistical analysis and training process of ML algorithms, leading to lower accuracy. There are various visualization and mathematical techniques by which we can detect and remove outliers, such as boxplots, z-scores, and interquartile range (IQR). In the proposed model, the outliers were detected by IQR and were handled by a capping technique.

5.3.4.2 Feature Scaling

During the preprocessing, feature scaling (FS) also counts among the important steps. It is important to bring all the features into the same scale in order to enable the model to learn from all the features equally without leading to bias. FS also helps algorithms to achieve the minimal cost function. Scaling plays a very significant role in making the ML model either a weak model or a better one [16]. FS is achieved by adopting methods like normalization and standardization. In this study, the FS was done by normalization method, as depicted in Equation (5.1) [16].

$$X_{\text{norm}} = \frac{X - \min(X)}{\max(X) - \min(X)} \quad (5.1)$$

5.3.5 Model Building

The basic steps of building an ML model are as follows:

- i. Selection of the ML model
- ii. Fitting of the ML model
- iii. Validating ML model

Selecting an ML model is the process in which a single ML model is selected out of various possible candidates for training a dataset [17]. Fitting an ML model involves training it on a dataset to find the best parameters that capture patterns in the data. The model's parameters are adjusted to minimize the difference between predicted and actual values, aiming to create an accurate predictor for new data. The functionality of an ML model to predict values and perform classifications is achieved by generalizing the model to new datasets [18].

Validating an ML model is a crucial and important process that is performed after training a model, where the trained model is assessed by testing data. Validating the ML model's outcome is necessary to ensure its accuracy [18].

5.3.5.1 *Methods*

The technique proposed in this study is the ensemble technique (ET). The main aim of using ET in this study is to overcome the issues of simple modeling of one classifier and to achieve better accuracy [17]. The ET involves the integration of multiple base learners, where every base learner contributes to the outcome leading to the highest level of accuracy and reduced error of prediction. ET is particularly used to help with decision-making problems.

One type of ET is bagging, which follows the process of sub-sampling training data in order to improve the outcome of a single type of classifier. The method that is used to subsample the data is bootstrapping. The bootstrapping process involves taking samples with the replacement from a dataset and calculating a statistic (mean, median) for each sample. So this process helps to get a distribution of results for statistics of interest by which we can estimate the standard errors and confidence intervals, which eventually leads to getting a less biased estimate. The output gathers the prediction of every learner, and with the help of majority voting, the desired output is achieved [19]. Figure 5.4 depicts the ET technique used.

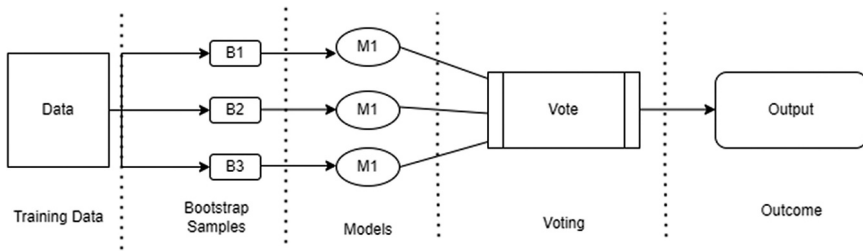


FIGURE 5.4
Bagging technique.

5.3.5.2 Technique

The first model used in this study is DT. The main reason for selecting this algorithm is that it can handle different types of data and does not require much data preprocessing [20]. Mostly, DT is used for classification tasks since DT simulates human-level thinking, which offers a straightforward way to analyze data and derive meaningful interpretations. This model involves building a treelike structure for the entire dataset and processing a single outcome at each leaf node [21]. The problems associated with training only one DT can be tackled by using the bagging model of DT, which leads to attaining better accuracy.

Deep learning is widely used in recommendation systems in order to enhance the sparse data and scalability of the method. This measures the enhancement in the quality of recommendation outcomes.

The second model trained in this study is the Neural Network (NN). The main criterion for selecting this algorithm is that it does a wondrous job of making the machines able to make decisions similar to the human brain and hence leads to better predictions by understanding the complex relationship between input and output data [22]. In this study, a homogenous bagging model was built.

5.3.5.2.1 Decision Tree

The DT algorithm is a straightforward as well as adequate supervised learning method in which the sample points are continuously divided depending on several parameters. The components of DT comprise the root node, some branches, and leaf nodes. The interior nodes in the structure trees provide descriptions related to different test scenarios.

This algorithm can be envisaged as a graphical treelike structure that predicts the output by employing a variety of specialized factors [21].

Figure 5.5 depicts the first technique. The working of the DT is illustrated in Algorithm 5.1.

Assume S to be a collection of training samples, each of which has a known class label. One feature is used to decide the class of training samples. Let's assume there are n classes. For $I = 1, \dots, n$, let S consist of s_i samples of class C_i for $I = 1, \dots, n$. A random sample has a probability of s_i/s that it belongs to the class C_i , where s is the number of total samples in the collection S . The anticipated information required to classify a given sample [3] is represented in Equation (5.2) [3].

$$I(s, s_1, \dots, s_n) = - \sum_{i=1}^n \frac{s_i}{s} \log \frac{s_i}{s} \quad (5.2)$$

Feature B with values $\{b_1, b_2, \dots, b_v\}$ can be employed to split S into the subsets $\{S_1, S_2, \dots, S_v\}$, where S_j consists of those samples in S that have a value b_j of B . Let S_j consist of s_{ij} examples of class C_i . The entropy of B is the predicted information based on this division by B . It is the weighted average that is represented in Equation (5.3) [3]:

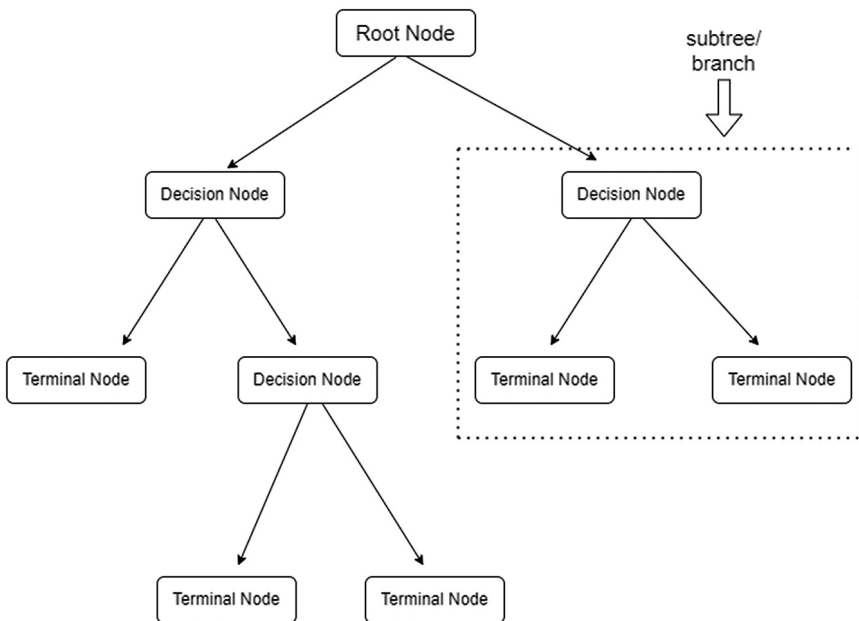


FIGURE 5.5
Decision tree.

ALGORITHM 5.1 DECISION TREE**Input** – Supervised Data of N instances

For each sample in the dataset:

- 1: Examine each base scenario.
- 2: For every feature b : evaluate the standardized information gained from splitting on b .
- 3: Let b_best be the feature with higher standardized information gain.
- 4: Build a decision *node* that splits on b_best .
- 5: Recurse on the subtree attained by splitting on b_best , and add those nodes as leaf nodes.

Return Classifier

$$E(B) = \sum_{j=1}^v \frac{S^1_j + \dots + S_n_j}{S} I(S_j + \dots + S_n_j) \quad (5.3)$$

This portioning on B results in an information gain that is represented in Equation (5.4) [3].

$$\text{Gain}(B) = I(s, s, \dots, S_n) - E(B) \quad (5.4)$$

For the relevance analysis, the information gain is computed for every attribute present in sample S . The attribute that exhibits the greatest information gain is regarded as the most distinguishing characteristic within the set. By calculating the information gain for each attribute, a ranking of the attributes is achieved, which can be used for relevance analysis to select the attributes to be used in concept description.

5.3.5.2.2 Neural Network

The nonlinear mathematical model created by mocking the biological brain is known as NN or ANN. ANNs are capable of processing nonlinear datasets and mapping them to the results. The most popular ANN for solving nonlinear datasets is multilayer perceptron (MLP). ANN consists of three main layers: input layer, hidden layer, and output layer. They are

built on a network of linked nodes known as neurons. Following that, connections between these neurons are used to send signals. As learning progresses, the weights assigned to the neurons and connections are updated and modified, which boosts their performance [23], as shown in Figure 5.6. NN presents the second technique. The working of the DT is illustrated in Algorithm 5.2.

Each neuron in the receiving layer (input layer) passes information and is transmitted to connected neurons in the hidden layer. Each hidden layer unit computes the output by adding the afflicted input signals that are employed in the activation function (AF) by applying Equation (5.5) [24].

$$h(t) = f\left(\sum x(t)W_H + B_H\right) \quad (5.5)$$

The output signal is calculated by adding the input signal and its AF by using Equation (5.6) [24].

$$O(t) = f\left(\sum h(t)W_o + B_o\right) \quad (5.6)$$

AFs are employed to each neuron present in the network. It is a mathematical “gate” between the receiver layer serving the current neuron and its result passing to the next layer. AFs can be linear and nonlinear.

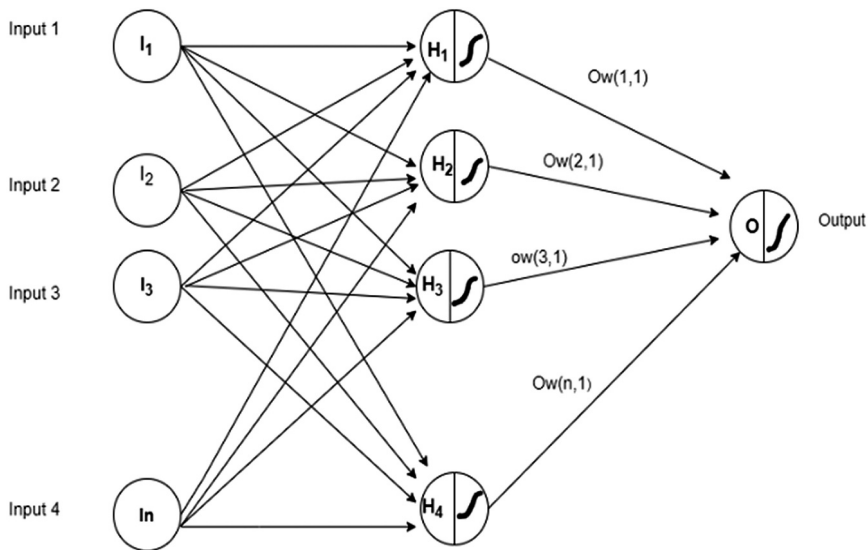


FIGURE 5.6
Neural network.

ALGORITHM 5.2 NN**Input** – Supervised Data of N instances

For each sample in the dataset:

- 1: Initialization: Input each row of observations on the input layer and every feature at each input node. Initialize the weight values randomly but not 0.
- 2: Forward-Propagation: From left to right, compute and activate each node of layers with the weights controlling the impact of corresponding neurons.
- 3: Cost Computation: Measure the error with the predicted value and the actual value.
- 4: Back Propagation and Updates: From right to left, update the weights with the gradient descent to lower the error until optimization.
- 5: Prediction: Predict the value for test data.

Return Classifier

Nonlinear AFs are employed in the case of complex data to predict the outcome. Among the AFs, ReLU (Rectified Linear Unit) is a diversely employed activation function. The inputs with a value less than zero are mapped to zero and then mapped to one for the inputs with a value greater than zero. The best option for multiclass classification is the Softmax function, as it transforms the results into a normalized probability distribution. The outcomes of the NN pass along to the Softmax (AF) that transforms the results into probabilistic values, which finally are compiled up to one.

Softmax output function can be expressed in Equation (5.7) [24]:

$$\sigma = \frac{e^{z_i}}{\sum_{j=1}^k e^{z_j}} \quad (5.7)$$

In feed-forward NN, back propagation is the most frequently employed algorithm for training the multilayer NN, where the signals from the input layer are transmitted onward, and errors are transmitted backward

to alter the weights in such a way that minimizes the output error. This process is replicated during training, and each iteration is known as an epoch. The error calculation in a back-propagation algorithm is presented in Equation (5.8) [24].

$$E = \frac{1}{2} \sum (t_k - O_k)^2 \quad (5.8)$$

Each weight is reconditioned as expressed in Equation (5.9) [25]:

$$wi = wi + \Delta wi, \quad (5.9)$$

where Δwi is the rectified factor calculated as represented in Equation (5.10) [25].

$$\Delta wi = \eta \delta j x_{ij} \quad (5.10)$$

5.3.6 Model Evaluation

The process of assessing the performance of a model is called model evaluation. Since predicting the best crop is a job that is affiliated with multiclass classification, there are four possible outputs – i.e., True Negative (TN), True Positive (TP), False Negative (FN), and False Positive (FP) [25].

For assessing the performance of a model, various metrics have been employed – i.e., accuracy, specificity, sensitivity, and area under receiver operator characteristic (AUROC) curve [26]. These metrics are defined in terms of the aforementioned results. To assess the stability and versatility of the proposed system, a method has been employed for validating the model, which is known as K-fold cross-validation (KFCV). The measures employed are defined as follows.

- Accuracy (Acc) = $\frac{TP + TN}{TP + TN + FP + FN}$
- Sensitivity (Sens) = $\frac{TP}{TP + FN}$
- Specificity (Spec) = $\frac{TN}{TN + FP}$
- F1 score = $\frac{2 * Precision * Recall}{Precision + Recall}$

5.3.6.1 AUROC Curve

An essential assessment statistic for classification problems is the AUROC curve. The AUROC curve is a crucial evaluation metric used to assess the performance of a multiclass classification problem. It is widely utilized to check and visualize how well a classification model performs. The curve successfully distinguishes between noise and signal by plotting True Positive Rate (TPR) vs. False Positive Rate (FPR) at various levels. Among the values, the value present at the top-left side of the curve is considered the optimum value. Figure 5.7 shows the AUROC curve [26].

5.3.6.2 K-Fold Cross-Validation

The process of assessing the ML models on small samples of data is known as cross-validation. The technique utilizes a parameter called k , which determines the number of groups into which the data has been divided. For every iteration, $k-1$ subsamples are employed for training the model, and the residual part is employed for assessing the performance of the model. Finally, the results from k -iterations are added, and the model's

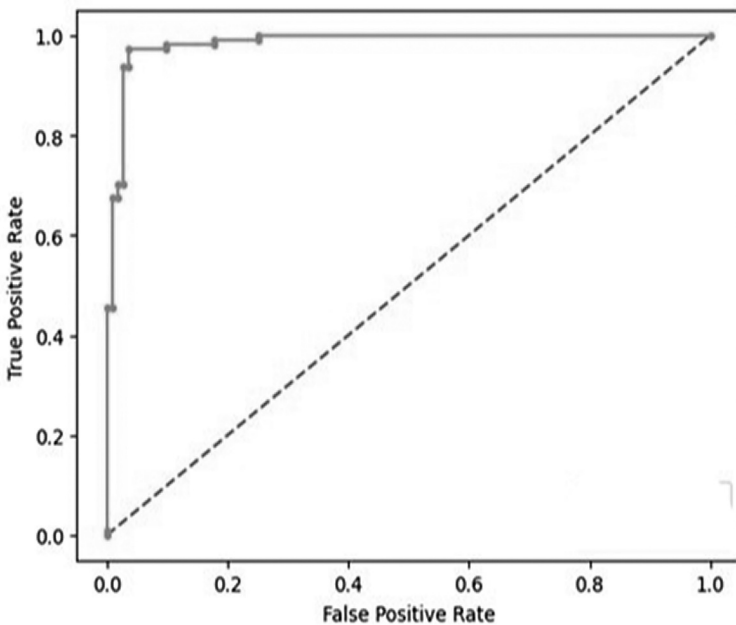


FIGURE 5.7
AUROC curve.

TABLE 5.2
Evaluation Parameters

Algorithm	Classification			
	Accuracy (%)	Precision	Recall	F1 Score
Decision Tree	0.98	0.96	0.98	0.96
Bagging Decision Tree	0.99	0.98	0.98	0.98
Neural Network	0.96	0.97	0.97	0.97
Bagging Neural Network	0.99	0.98	0.98	0.98

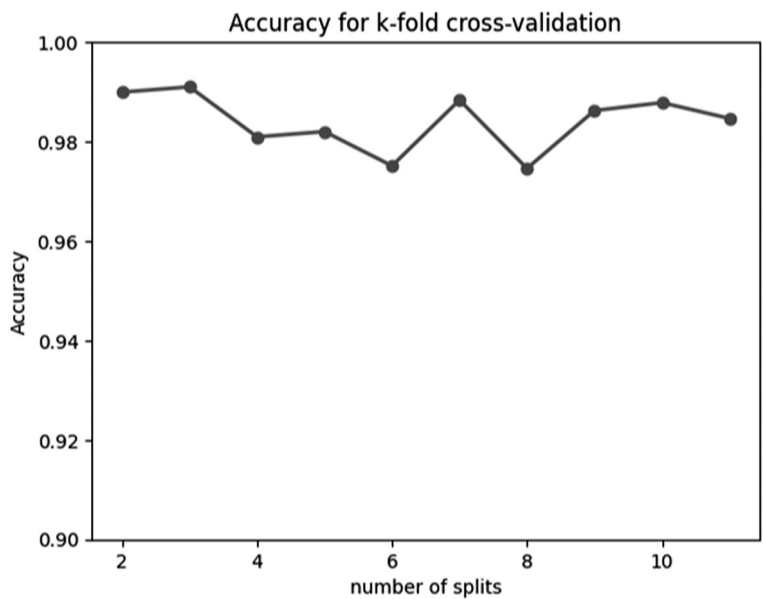


FIGURE 5.8
KFCV.

mean accuracy is generated from the average results, as shown in Table 5.2 [27]. Figure 5.8 depicts the accuracy attained in each iteration of the KFCV technique.

5.4 RESULTS

In the proposed system, the results of the model are explained in reference to assessing parameters. In order to analyze the effectiveness of the system, the KFCV data is also provided [28–32]. The model attained

accuracy, sensitivity, and specificity of 0.99, 0.98, and 0.98, respectively, and an AUROC value of 0.998, which is considered among the good results. Another measure to check the authenticity of the model, nine-fold cross-validation, has been executed. The dataset was divided into nine folds, eight of which were employed for training the ML model, and the remaining one fold was set aside for assessing the model.

5.5 CONCLUSION

To date, the yield obtained from crop production is inadequate due to insufficient understanding of the factors influencing production and reliance on conventional farming methods. The recommended system based on ensemble learning will help the agricultural community overcome the corresponding challenges involved. Ultimately, this will contribute to an increase in crop production, leading to higher yields. The proposed model attained sensitivity, specificity, accuracy, and AUROC of 0.98, 0.98, 0.99, and 0.998, respectively. Among the proposed models, the ensemble models outperformed the other two single-classifier models [33–37]. The proposed model will save the efforts of agriculturists required to select the best crop, leading to higher yields automatically. However, it is relevant to indicate that this model is based on finite parameters. However, there are several issues that can be addressed in the future, such as taking other parameters into consideration for model training and tuning other parameters of the proposed models.

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